IMPACT-Scribe

User Study Guide

Interactive Action Segmentation Review Tool

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1. Overview

IMPACT-Scribe is an interactive action segmentation review tool. Given a video and an initial (machine-generated) action segmentation, you will review and correct the segmentation by:

- Following the system's query suggestions (boundary or label questions)
- Drawing temporal scribbles to refine boundaries
- Accepting or rejecting proposed corrections

Your goal is to produce a high-quality action segmentation with minimal effort. The tool learns from your corrections and improves its suggestions over time.

2. Getting Started

Step 1: Load a Video or Baseline

Click the menu button in the top-left toolbar and choose "Open Session". Select a video file and, optionally, an existing annotation JSON. The system loads the video and displays the baseline segmentation on the timeline.

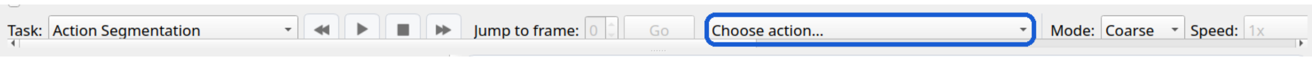

Fig 1. Load a video and baseline segmentation

Step 2: Inspect the Workspace

After loading, the workspace shows the video player on the left and the action timeline on the right. Each colored segment represents one action label. Hover over the timeline to preview any frame.

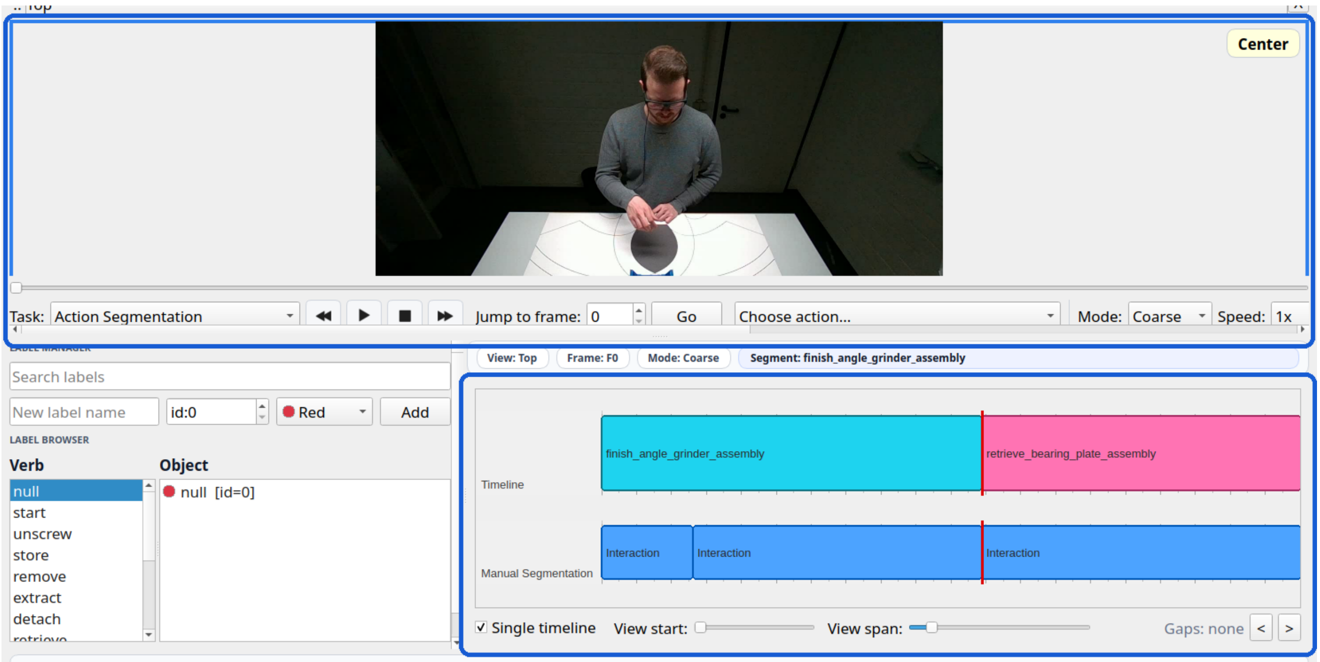

Fig 2. Workspace with video and timeline

Step 3: Click Suggest Query

Click the "Suggest Query" button in the toolbar. The system analyzes the current segmentation and suggests the most valuable boundary or label to review next. A suggestion card appears in the footer area.

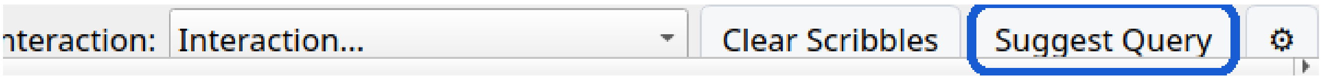

Fig 3. Suggest Query generates a review target

Step 4: Review the Suggestion

The footer card shows the suggested question: a boundary to check or a label to verify. Read the description and decide whether to accept, reject, or refine with a scribble.

Fig 4. Review the suggested boundary or label

Step 5: Refine and Accept

If the boundary needs adjustment, draw an uncertain scribble stroke across the suspicious region on the timeline. The system proposes a refined boundary (red line). Drag to fine-tune, then click Accept.

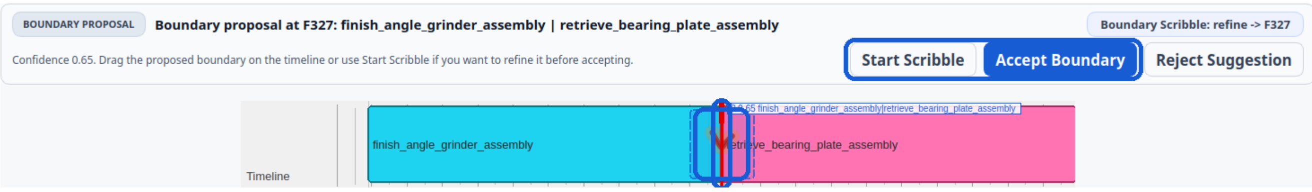

Fig 5. Draw a scribble, refine, and accept

3. Core Workflows

3.1 Direct Timeline Annotation (Basic)

The simplest way to annotate is to draw segments directly on the timeline. No special mode is required - this works at any time:

- 1. Select a label from the Label Panel on the left (click a Verb, then an Object)
- 2. On the timeline, click and drag in an empty area to draw a new segment. The segment appears as a colored block matching the selected label.
- 3. To resize: drag the left or right edge of any existing segment
- 4. To move: drag the center of an existing segment
- 5. To delete: right-click on a segment to remove it
- 6. Ctrl+Click on a segment to split it at that frame

This is useful when you want to annotate from scratch or quickly fix a region without using the query planner.

3.2 Query Suggestion Workflow (Guided)

For efficient review, the query planner helps you focus on the most valuable corrections. It selects the next question based on:

- Boundary uncertainty: where the model is least confident
- Label disagreement: where features suggest a different label
- Multi-view conflict: where different camera views disagree
- State conflict: where action labels violate assembly rules

After clicking Suggest Query, three actions are available:

- **Accept Suggestion** - Apply the proposed change directly
- **Start Scribble** - Enter scribble mode to refine the boundary first
- **Reject Suggestion** - Skip this query and move on

Repeat: click Suggest Query again after each decision. The planner adapts based on your corrections.

3.3 Temporal Scribble Interaction

Scribbles are short strokes you draw on the timeline in Boundary Scribble mode to tell the system where a boundary might be. There are three scribble types:

- **Uncertain (default)** - "I think a boundary is somewhere in this range"
- **Left** - "The left side of this region belongs to this label"
- **Right** - "The right side of this region belongs to this label"

Scribble workflow:

- 1. Enter scribble mode via the Interaction dropdown or "Start Scribble" button
- 2. Click and drag on the timeline across the suspicious boundary region
- 3. The system proposes a boundary split (red marker) with left/right labels

- 4. Optionally drag the red marker to fine-tune the exact position
- 5. Click Accept to apply, or Reject to discard
- 6. Click "Clear Scribbles" to reset the current scribble session

3.4 Manual Global Segmentation

Select "Manual Segmentation" from the Interaction dropdown. This activates a full manual mode where segment boundaries are placed frame by frame. Use this for precise control when the automated workflow is not needed.

4. Interface Reference

4.1 Toolbar Buttons

- **Play / Pause** - Toggle video playback
- **<< / >>** - Step backward / forward by 10 frames
- **ASOT Pre-label** - Generate a machine baseline segmentation
- **Magnifier** - Toggle zoom selection on the video
- **Validation** - Toggle validation overlay on/off
- **Interaction dropdown** - Switch between Boundary Scribble / Manual Segmentation / Exit
- **Clear Scribbles** - Reset current scribble strokes and proposals
- **Suggest Query** - Ask the planner for the next review target
- **Settings** - Open settings dialog (Ctrl+,)
- **Quick Start** - Open the in-app quick start guide
- **+ Add View** - Add an additional camera view

4.2 Timeline

The timeline is the primary annotation area:

- Colored blocks = action segments with label names
- Red vertical line = current playhead position
- Hover = preview that frame in the video player
- Dotted lines = segment boundaries (draggable)
- Red marker = proposed boundary from scribble refinement
- Scroll wheel = pan the timeline left/right

Mouse actions on the timeline:

- Left click + drag on empty space = create a new segment (uses selected label)
- Left drag on segment edge = resize segment boundary
- Left drag on segment center = move the entire segment
- Right-click on segment = delete it
- Ctrl + click on segment = split at that frame

4.3 Label Panel

The label panel (left side) shows all available action labels organized by verb-object structure. Click a verb to filter, then click an object to select the label for annotation.

- Search box: filter labels by name
- Add button: create a new label with name, ID, and color
- Double-click: rename an existing label inline

4.4 Footer / Query Card

When a query is active, the footer shows:

- Query type: BOUNDARY SUGGESTION or LABEL SUGGESTION

- Target frame range and labels involved
- Confidence score
- Action buttons: Accept / Reject / Start Scribble

5. Keyboard Shortcuts

Navigation

Space	Play / Pause
A	Step back 1 frame
D	Step forward 1 frame
Shift+A	Step back 10 frames
Shift+D	Step forward 10 frames
J	Seek back 1 second
K	Pause
L	Seek forward 1 second
Home	Jump to start
End	Jump to end

Editing

Ctrl+Z	Undo
Ctrl+Y	Redo
Ctrl+,	Open Settings

Assembly State Mode (PSR)

Ctrl+K	Split segment at playhead
Ctrl+Shift+S	Set scope to current segment
Ctrl+Shift+F	Set scope from current frame forward
Ctrl+Backspace	Reset selected segment
Ctrl+I	Invert selected segment state
Ctrl+M	Merge adjacent identical states

Video Player

Ctrl+Scroll	Zoom in / out on video
Left Drag	Pan zoomed video view
Double Click	Reset zoom to fit window

6. Multi-View Support

IMPACT-Scribe supports up to 5 synchronized camera views. Click "+ Add View" to load an additional video. All views share the same timeline and are synchronized frame-by-frame.

- Hovering or seeking in one view updates all views
- Labels can differ between views (per-view annotation)
- The query planner considers cross-view disagreement

7. Import / Export

- **Open Session** - Load video + optional label map + annotation JSON
- **Import JSON** - Load annotation data for selected views
- **Export JSON** - Save all annotations to a single JSON file
- **Export per-view** - Save separate JSON files per camera view
- **Import/Export Label Map** - Load or save the label vocabulary (TXT)

8. User Study Task Instructions

8.1 Your Task

You will be given a video with a machine-generated action segmentation baseline. Your task is to review and correct this segmentation using the tool's guided workflow.

8.2 Procedure

- 1. The system will load a pre-configured session for you
- 2. Click "Suggest Query" to receive the first review question
- 3. For each suggestion, decide: Accept, Reject, or Refine with Scribble
- 4. After each decision, click "Suggest Query" again
- 5. Continue until you are satisfied with the segmentation quality
- 6. Export the final annotation when done

8.3 Tips

- You can always draw segments directly on the timeline without any special mode
- Trust the query planner: it prioritizes the most impactful corrections first
- Use scribbles when a boundary is close but not exact
- Use Accept when the suggestion looks correct
- Use Reject to skip suggestions you disagree with
- Hover over the timeline to quickly preview frames before editing
- Use A/D keys for precise frame-by-frame inspection
- Ctrl+Z to undo any mistake

8.4 What Counts as Done

There is no fixed number of corrections required. You are done when you believe the segmentation accurately reflects the actions in the video. The system will record all your interactions for analysis.

9. Troubleshooting

- **No suggestion available:** Make sure a video and baseline are loaded first.
- **Scribble not working:** Ensure you are in "Boundary Scribble" mode from the Interaction dropdown.
- **Video not playing:** Check that the video file path is accessible. Try re-loading.
- **Labels missing:** Import a label map TXT file via the menu.
- **Need help:** Click the Quick Start button (top-right) for the in-app guide.